



MAMDOUH ATTIA

Software Engineer

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PROFESSIONAL SUMMARY

Computer Engineering graduate (Cairo University) with **3+ years** building **cross-platform** mobile apps for real clients, open-source projects, systems, and AI-powered software. Delivered end-to-end solutions using **Flutter**, **Python**, and **C/C++**. Strong foundation in systems programming, **data structures**, **algorithms**, and **ML**. **200+** hours technical instruction across **ML**, **Python**, **Embedded**, and **Flutter** programs.

EDUCATION

Cairo University — Faculty of Engineering, Computer Engineering

2019 – 2024

Grade: Very Good with Honors | Graduation Project Grade: Excellent (A+)

PROFESSIONAL EXPERIENCE

Flutter Developer — Freelance, PrimaLab, Itqan Community

Sep 2023 – Present

- Delivered +3 client Flutter apps (HyperMate, VolunTeen, Check-It) integrating Hardware IoT sensors (ESP8266), AI chatbots (GPT-3), Firebase, REST APIs, QR/barcode scanning.
- Contributed to open-source mushaf-imad-flutter package (PR #27); Integrating Quran.com's audio API : OAuth 2.0, hybrid audio engine, Stream Transformers, 80+ Mocktail tests, 86% latency reduction (by token caching).
- Applied MVC, MVVM, Clean Architecture; owned full SDLC from Figma wireframes to tested releases.

Technical Instructor — DEPI/MCIT @ Harvest, DECI/MCIT Eyouth, Maspero Academy, Mindset, PrimaLab

Aug 2023 – Present

- DEPI Initiative:** Led 100+ hrs ML Engineering track for 15+ engineers (From Maths, Python to Azure AI); designed & graded 17 tasks and final capstone. **Maspero Academy:** Delivered 20+ hrs Python basics for 35 students.
- PrimaLab:** 50+ hrs Dart/Flutter across 28 sessions for 10+ students. | **Mindset:** Taught System programming – Embedded systems – AI.

Flutter Developer Intern — Orange Digital Center Software House

Jun 2023 – Sep 2023

- Worked in Agile squads on 2 internal mobile apps and 1 hackathon e-wallet. Translated Figma designs to Flutter UI using MVVM + Bloc/Cubit.

IT Support Specialist — Future Of Egypt (Mandatory Military Service)

Dec 2024 – Nov 2025

- Technical support for hardware, software, and networks in a government operational environment.

SELECTED PROJECTS

Mushaf-IMAD Package | Dart, REST API, OAuth 2.0, Streams, Mocktail

Feb – Mar 2026

- 86% latency reduction via Single-Trip optimization; hybrid timing hierarchy (RAM → Local → API) for zero-latency audio playback sync.
- +70 atomic commits of 5000+ lines, 80+ async Mocktail tests; published architectural guides for open-source maintainability.

Predicto (Graduation Project, A+) | Python, PyTorch, OpenCV, Desktop GUI

Aug 2023 – Jul 2024

- Team lead: built vPredicto open-source library for Video Frame Prediction with full PyQt desktop app (model selection, video input, output).
- Implemented from scratch:** *SimVP* [co-implemented] (CNN Encoder-Translator-Decoder, TPU-optimized, 5x training speedup via checkpointing) and PredNet (ConvLSTM predictive coding, resolved critical dimension-mismatch bugs).

VolunTeen | Flutter, Firebase, GPT-3, Provider, MVC

Sep 2023 – Jan 2024

- Solo developer for real client: 30–40 responsive screens, GPT-3 chatbot, Firebase storage, in-app video player, WhatsApp/URL launcher.

HyperMate | Flutter, C, ESP8266, Wi-Fi, GPT-3

Nov – Dec 2023

- IoT stress-monitoring app with real-time GSR sensor readings over Wi-Fi, personalized exercise, AI chatbot. Also authored Arduino firmware.

Distributed File System (Go-DFS) | Go, gRPC, Protocol Buffers, TCP

Mar – Apr 2024

- Master-Worker architecture: gRPC heartbeat monitoring, auto-replication engine (factor 3), Data Keeper self-scanning, and MP4 upload/download pipelines via raw TCP.

OS Scheduler | C, Linux, IPC, Memory Management

Apr 2022

- Led Process Generator: fork()/exec() lifecycle, IPC Message Queues. Designed Buddy Allocator (1024-byte pool, recursive splitting/merging, allocate() + deallocate() functions).

Gamingo Engine + Drive Me Home Game | C++, OpenGL, GLSL, OOP

Mar – May 2023

- Custom 3D game engine: Role: Shader, Texture, Material, Sampler classes + GLSL programmable shading. Collision detection, game-state machine, post-processing shaders (Sepia, CRT, Inversion).

MIPS Processor | Verilog, Computer Architecture, FPGA Simulation, Modelsim

Dec 2022

- 5-stage pipelined MIPS processor. Role: decode stage, top-level integration (5-stage buffers), control unit ISA mapping, hazard detection and forwarding verification.

SKILLS

Programming:	Dart · C/C++ · Python · Java · Go (basic) · SQL	Web (basic) :	HTML · CSS · JS · Node.js · Database
Mobile / Frontend:	Flutter · Provider · Bloc/Cubit · MVC · MVVM · Clean Architecture · Streams · BAAS (Firebase · Supabase) · REST APIs		
AI / ML:	PyTorch · TensorFlow · scikit-learn · OpenCV · Computer Vision · NLP · pandas · numpy · Machine Intelligence · ML · DL		
Systems:	Linux · Git/GitHub · GitHub Actions · Algorithms · Data Structures · Design Patterns · OOP · SOLID SDLC · Agile/SCRUM · gRPC		
Hardware:	ESP8266 · Wi-Fi · Arduino Firmware · Hardware Sensors · Embedded Systems · IC Design · Verilog · VLSI		

CERTIFICATES

GitHub Professional Certificate (2026)

Flutter Internship — ODC (2023) ·

Machine Learning Specialization — Coursera (2024)

HCIA-AI — Huawei International (2021)